

1. Introduction

What is EEG-based Motor Imagery?

Electroencephalography-based motor imagery (EEG-MI) is a technology that reads brain signals when a person imagines moving, allowing those signals to be translated into commands for electronic devices (Wolpaw et al., 2002). It can be used to help people control wheelchairs, robotic arms, and computer interfaces.

However, these brain signals vary from person to person and can also change over time (Lotte et al., 2018), so each MI session usually needs a calibration step before it can work properly.



Source: The University of Texas at Austin News

The Traditional Calibration Method

Calibration is the process where the system collects brain signals from the user while they perform specific tasks, then learns from those signals. The Graz method asks users to follow repeated on-screen prompts and imagine the same movement many times during each session. This method is still effective, but it usually takes 15–30 minutes per session, requires expert supervision, and can be tiring (Giles et al., 2022). As a result, it may reduce user engagement and lower data quality over time (Lotte et al., 2018).

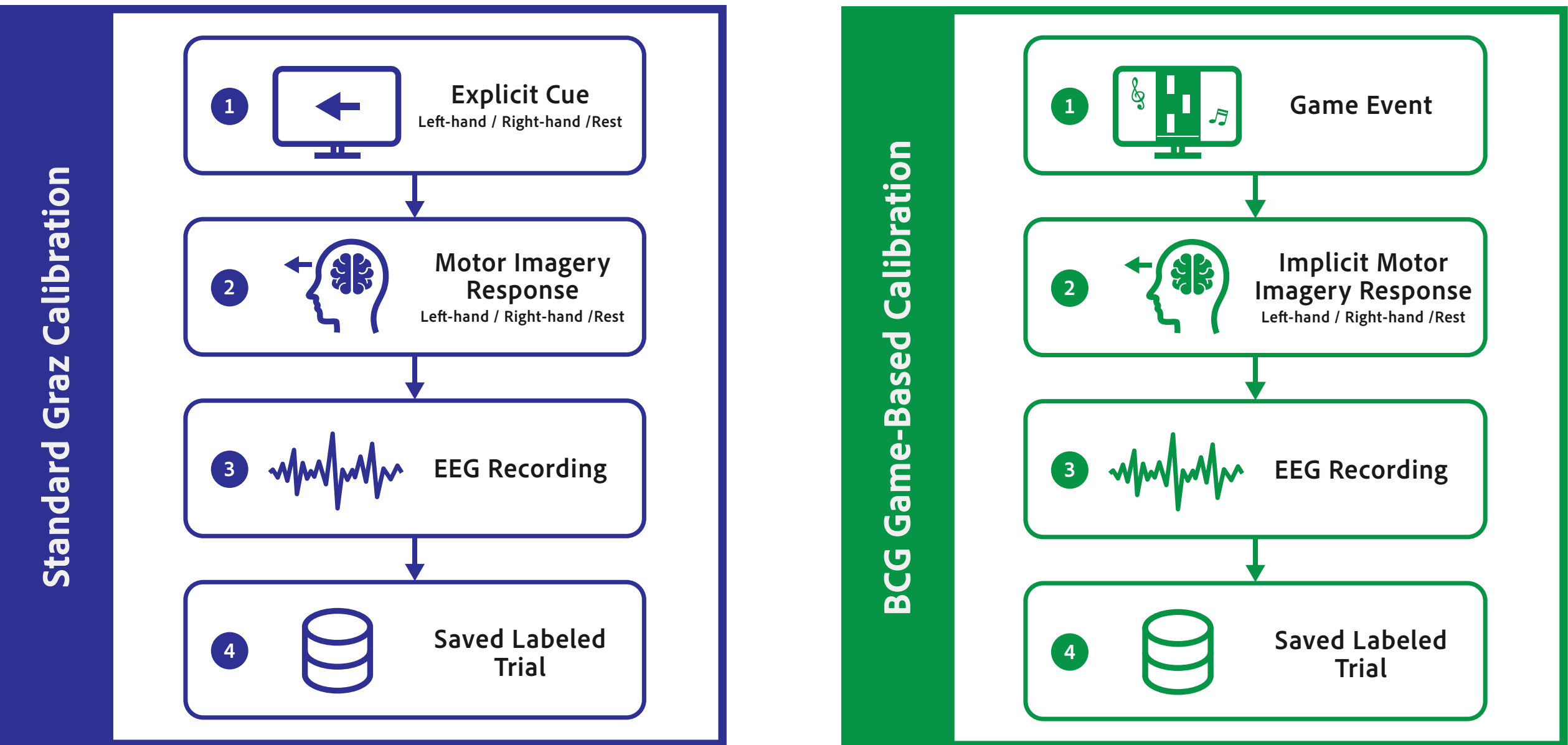
2. Objectives

- 1 Develop a dual-mode BCI software calibration framework (traditional Graz and gamified protocols)
- 2 Evaluate user experience by comparing fatigue and workload using NASA-TLX.
- 3 Establish a hardware-compatible baseline classifier for 14-channel motor imagery.
- 4 Build an automated pipeline for data ingestion and personalized fine-tuning.

3. Methodology

BCG Concept Compared to Graz Method

In the standard Graz protocol, users follow explicit on-screen cues to repeatedly imagine left-hand, right-hand, or rest states, producing clearly labeled EEG trials but often creating a repetitive and fatiguing experience. In contrast, BCG embeds the same calibration targets within gameplay events, so the user performs the required motor imagery tasks more naturally through interaction with the game. This design aims to preserve structured data collection while reducing boredom and mental strain, enabling labeled calibration data to be gathered implicitly rather than through direct commands.

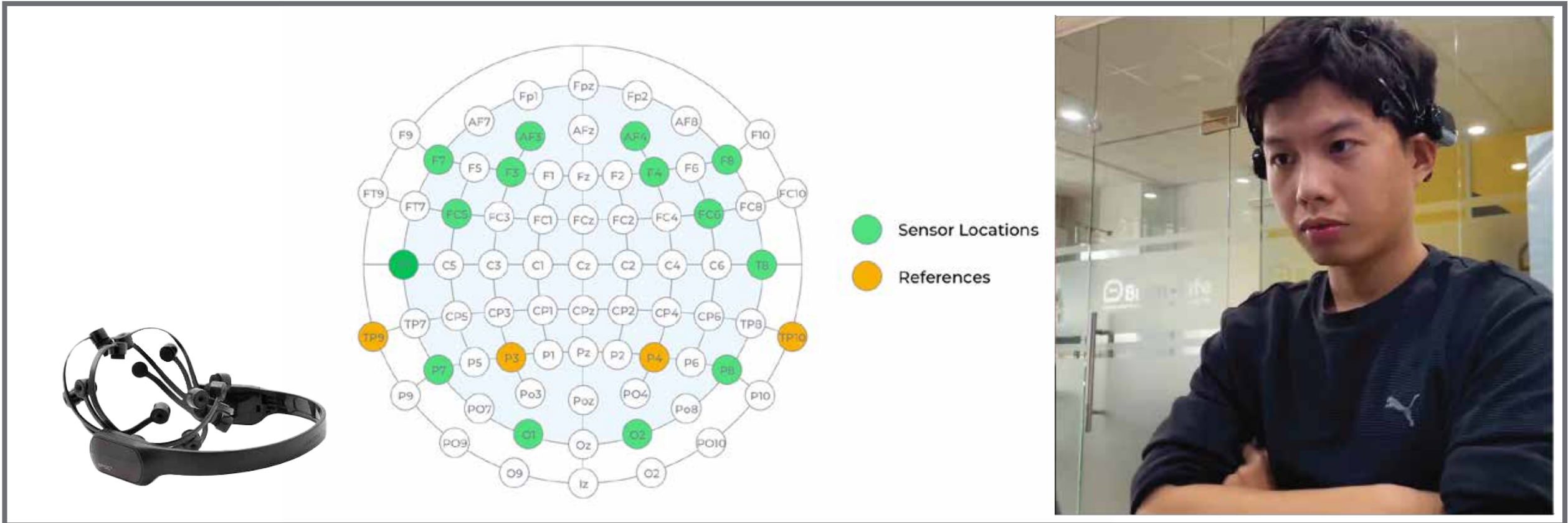


BCG Game Mechanic Design

When designing BCG, we framed it as an MI-as-data-collection game, where the primary goal is to collect correctly labeled EEG data for calibration rather than to enable free player control. The game offers only two choices at each decision point, corresponding to left-hand and right-hand motor imagery, while rest trials are recorded in the intervals between them. This ensures that each in-game event maps clearly to a motor imagery class and supports covert cueing instead of explicit instructions, as in the Graz protocol.

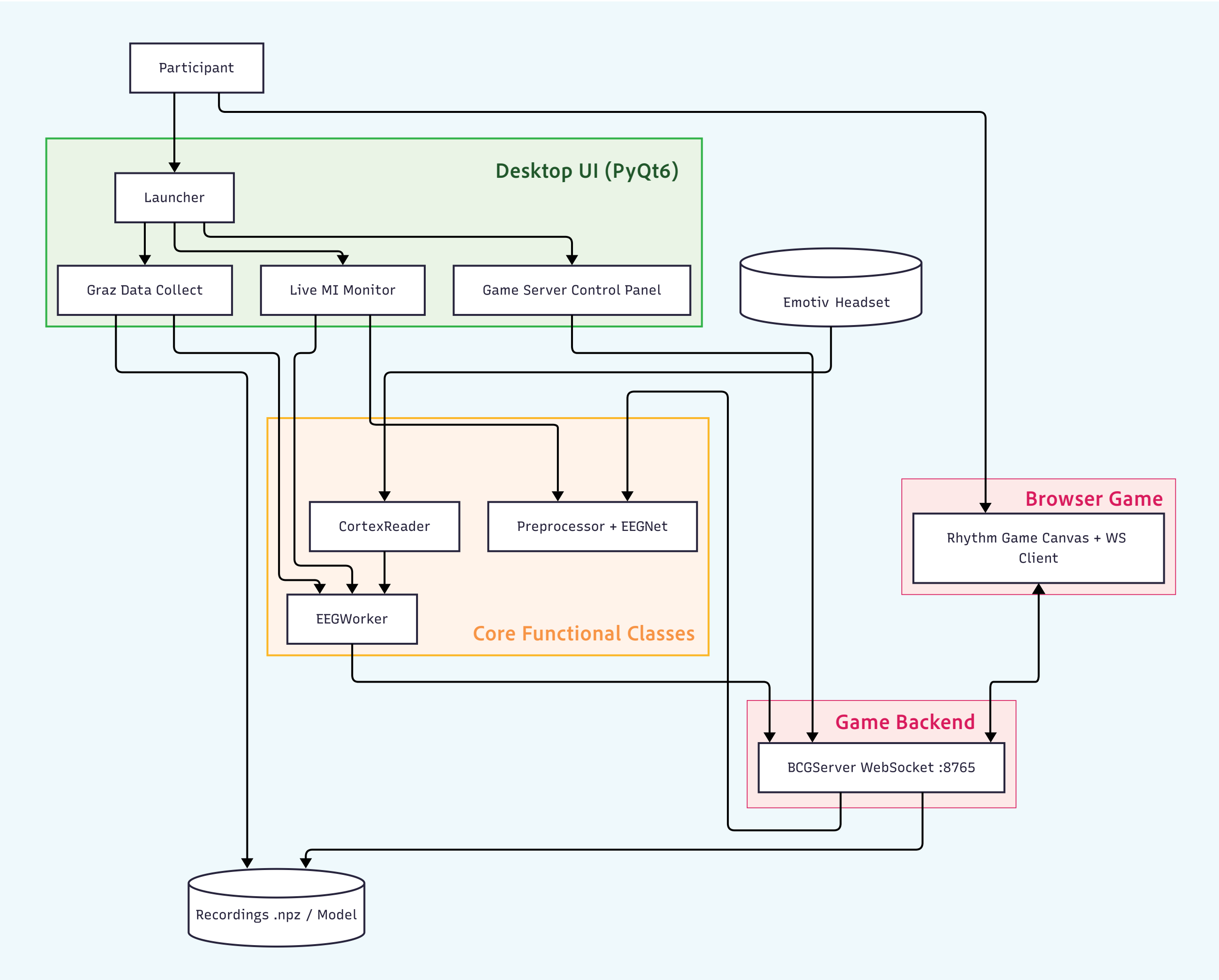
Device Selection and Setup

The experimental device used in this project is the Emotiv EPOC X headset. It provides 14 EEG channels placed according to the standard 10–20 system, including AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4. It also supports real-time signal acquisition with wireless connectivity and contact quality monitoring. Emotiv’s Cortex API is used to connect the headset to the computer and stream raw EEG data for recording and processing.



4. Artefact Design and Result

System Architecture

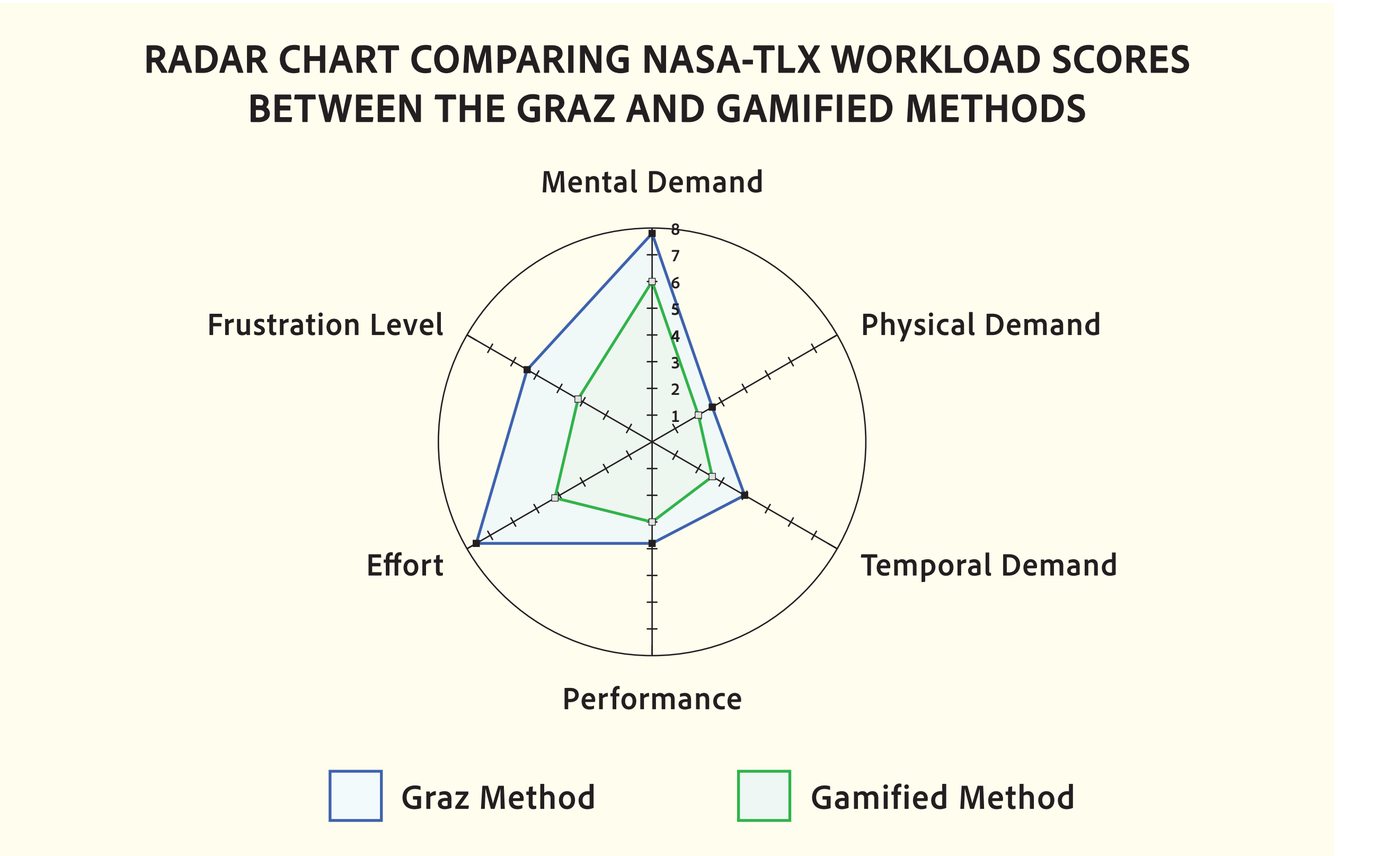


3. Evaluation

Evaluation Design

Between-subjects user study with **10** healthy participants to compare the Graz protocol and the gamified protocol. Participants were split into two groups, with 5 people in each condition. Each participant completed a 10-minute calibration task, and we used **NASA-TLX** to measure perceived workload and user experience.

User Study Result



4. Conclusion

- ¹ This study explored a gamified approach to BCI calibration and its impact on user workload and experience.
- ² The user study with 10 participants suggests that the gamified interface may reduce perceived effort, frustration, and mental demand compared with the traditional Graz method.
- ³ Overall, the findings indicate that gamification could help make motor imagery calibration more engaging and practical, supporting broader real-world adoption of motor imagery-based BCI systems.

References

Giles, J., Ang, K., Phua, K. and Arvaneh, M. (2022) 'A transfer learning algorithm to reduce brain–computer interface calibration time for long-term users', *Frontiers in Neuroergonomics*, 3, p. 837307.

Lotte, F., Bougrain, L., Cichocki, A., Clerc, M., Congedo, M., Rakotomamonjy, A. and Yger, F. (2018) 'A review of classification algorithms for EEG-based brain–computer interfaces: a 10 year update', *Journal of Neural Engineering*, 15(3), p. 031005.

Wolpaw, J., Birbaumer, N., McFarland, D., Pfurtscheller, G. and Vaughan, T. (2002) 'Brain–computer interfaces for communication and control', *Clinical Neurophysiology*, 113(6), pp. 767–791.